Milestone B: UI/UX Improvements - Delivery Report

Date: 18 January 2026

Preview URL: https://science-canvas-creator.vercel.app

Work Completed

1. Toolbar Consolidation & Reorganization

Simplified the left toolbar by grouping related tools into organized menus

- Combined 7 separate toolbar buttons into 3 organized dropdown menus
- Added clear section titles to all dropdown menus for easier navigation
- Reorganized toolbar layout for more logical tool grouping

New Grouped Tool Menus:

- **Lines and Connectors** - All line drawing tools in one menu
- **Parenthesis and Brackets** - All bracket/parenthesis shapes in one menu
- **Image Tools** - Image and Image Placeholder combined into one menu
- **Shapes Menu** - Now includes basic shapes, labeled shapes, and flowchart shapes
- **Annotation Tools** - Now includes zoom callouts

Other Toolbar Improvements:

- Moved Eraser tool from top toolbar to left toolbar for better organization
- Removed "Saving..." text indicators from toolbars
- Updated save button to show saving state (with spinner) and briefly display "Saved" confirmation
- Made all toolbar buttons uniform size so they don't compress on small screens
- Removed visual dividers between toolbar sections for cleaner look
- Added smart scrolling: chevron buttons appear when toolbar is too tall, scroll on hover

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2. Smart Button States

Buttons now enable/disable automatically based on what's selected

Prevents errors by disabling buttons when they can't be used:

- **Cut/Copy** - Only enabled when something is selected
- **Paste** - Only enabled when clipboard has content
- **Text Alignment** - Only enabled when text is selected
- **Arrange Menu** - Only enabled when something is selected
- **Group** - Only enabled when 2+ items are selected
- **Ungroup** - Only enabled when a group is selected

Tooltips still appear when hovering disabled buttons to explain what they do.

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3. Text Alignment Fix

Fixed text alignment buttons to work correctly

Previously, the align buttons in the top toolbar would move text objects around the canvas instead of aligning the text inside them.

Changes made:

- Top toolbar align buttons now format text (left/center/right) correctly
- Only enabled when text is selected
- Works the same as the floating text toolbar
- Removed alignment section from Arrange panel in right sidebar

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4. Premium Features UI Redesign

Streamlined access to premium features with clear visual feedback

Reorganized the header to make premium features more discoverable and provide clear progress indicators:

New Header Layout:

- Replaced the "{X}/3 Projects" text button with three dedicated icon buttons
- **AI Icon Generator button** - Opens AI icon creation dialog when unlocked
- **PowerPoint Generator button** - Converts diagrams to PowerPoint when unlocked
- **Share button** - Publishes current project for admin review (always available)

Locked State (when user has fewer than 3 approved projects):

- AI Icon and PowerPoint buttons display with subtle locked styling

- Clicking locked buttons opens an informative popover showing:
    - Progress bars for published and approved projects
    - Brief descriptions of premium features
    - Quick access to Share button to publish more projects
    - Typical review timeline (24-48 hours)

Unlocked State (3 or more approved projects):

- Locked styling removed from premium buttons
- All features become fully accessible
- Buttons directly open their respective tools

This design makes the path to unlocking premium features clear and provides immediate feedback on progress toward that goal.

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5. Icons & Assets Panel Redesign

Unified interface design with improved organization and consistency

Redesigned both the Icons and Assets panels to provide a consistent, intuitive experience for managing visual elements:

Icons Panel Reorganization:

- Added tab system: **All** and **Community** tabs
- **All tab** displays comprehensive icon library with organized categories:
    - "My Icons" category at top - shows icons saved to personal library from AI generator
    - Biosketch category with distinctive purple styling - featured scientific icons
    - All other icon categories below
- **Community tab** shows user-submitted icons:
    - "My Shared Icons" - icons the current user has submitted and been approved
    - "Community Icons" - AI-generated icons from other users after admin approval
- Added folder and user icons to tab labels for visual clarity

Assets Panel Updates:

- Matched thumbnail grid to Icons panel (4 columns instead of 5)
- Added hover preview cards showing larger thumbnails with file details
- Reduced thumbnail padding for more efficient use of space
- Added three-dot menu to all thumbnails with context-aware options:
    - My Files tab: Share/Unshare and Delete

    - Community tab: Unshare and Delete (only on your own shared assets)
- Replaced browser confirm dialog with proper confirmation dialog component
- Added primary theme color to Upload button border and icon

Improved Consistency:

- Both panels now use the same visual language and interaction patterns
- Tab styling matches between Icons and Assets panels
- Thumbnail sizing and hover states unified across both panels
- Menu interactions consistent throughout

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6. Properties Panel Redesign

Consistent styling and improved usability across all property sections

Redesigned the properties panel for better organization and user experience:

Redesigned tabs and sidebar

- Applied a consistent primary tab navigation and sidebar collapse styling
- Combined the "Properties" and "Styles" secondary tabs into just the "Styles" tab for simplicity and design consistency

Collapsible Sections:

- All sections now use consistent collapsible headers with chevron indicators
- Sections remember their collapsed/expanded state throughout the session
- Tool-specific sections were moved to the top, with added icons to visually distinguish them from generic sections

Fixed UI Issues:

- Background gradient controls now use the same sophisticated interface as object gradients
- Color pickers only appear when their respective checkboxes are enabled
- Color swatches (presets and recent colors) moved next to each color picker for better context

Style Improvements:

- Style preview cards show actual styling (dotted lines, bold text, colored shapes)
- Gradient preview shows live updates as settings change
- All sections use consistent spacing, borders, and typography

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7. Testing & Quality Assurance

All milestone tests completed and passing

Comprehensive testing performed across all deliverables:

Additional E2E Test Coverage:

- Save button state changes (enabled → saving → saved)
- Premium features UI (locked/unlocked states, popover behavior)
- AI Generated Icons section (button states, tab switching)
- Smart button states (cut/copy/paste, group/ungroup)
- Text alignment functionality

E2E coverage fixes

- Redesigning the UI required the existing E2E tests to be rewritten
- All tests were fixed to verify the functionality using the updated UI
- 100% pass rate maintained across all tests

Manual Testing:

- Properties panel organization verified across all tabs
- Toolbar consolidation tested with all tool groups
- Icon and asset panel consistency validated
- Premium feature progress tracking confirmed
- All UI interactions tested for proper behavior

All tests outlined in the milestone requirements have been executed and verified. The application performs as expected across all new and updated features.

Summary

✅ **Cleaner Toolbar** - 7 separate buttons combined into 3 organized menus

✅ **Better Organization** - Related tools grouped logically with clear labels

✅ **Improved Scrolling** - Hover-activated chevrons for smoother navigation

✅ **Premium Features Redesign** - Clear visual feedback and progress tracking for unlocking features

✅ **Unified Panel Design** - Icons and Assets panels now share consistent interface and interactions

✅ **Better Icon Organization** - Icons separated into All and Community tabs with featured categories at top

✅ **Properties Panel Redesign** - Consistent collapsible sections with persistent states and improved styling

✅ **Milestone Testing** - Additional milestone tests completed and passing

✅ **Comprehensive Testing** - All tests across app-wide functionality updated and passing

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